

Table 1. Key Assumptions of the Accelerated Failure Time (AFT) Model

Assumption	Description
Parametric Distribution	The survival time follows a specified distribution (e.g., Weibull, log-normal, or log-logistic).
Log-Linearity	Covariates exert a linear effect on the logarithm of survival time. Deviations from linearity are not modeled.
Independence of Observations	Each subject's survival time is assumed independent. If clustering exists (e.g., by site or sex), it is not modeled.
Non-Informative Censoring	Censoring is independent of the true event time; i.e., the probability of being censored is unrelated to the survival time.
Correct Covariate Specification	Covariates are accurately measured and appropriately modeled (continuous, categorical, or time-varying).
Proper Handling of Ties	When event times coincide, small differences are assumed in continuous time; ties will be broken randomly.
Stationarity of Effects	Covariate effects on log(survival time) remain constant throughout follow-up. Departures from stationarity are not modeled.
Appropriateness of Distributional Tail	The chosen survival distribution should capture long-term survival behavior (e.g., log-logistic for heavy tails).
Adequate Sample Size	A sufficient number of observed events is required for stable estimation, especially when modeling rare events.
Model Fit Validation	Goodness-of-fit will be assessed via AIC/BIC, visual overlay of model-predicted versus Kaplan-Meier survival curves, and other diagnostic tools.

Note: AFT = Accelerated Failure Time; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion. All analyses assume right-censoring and are modeled using the flexsurv package in R (v4.4.0).